

Biniyam Zergaw M.S. in Data Science

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SUMMARY

Software Engineer with 3+ years of experience in data analysis and software development, currently pursuing an M.S. in Data Science at Fordham. Experienced with Machine Learning algorithms, GCP, backend development with FastAPI using ORM, Pydantic, and frontend integration using React and Next.js. A strong collaborator focusing on performance, reliability, and intuitive user experience.

WORK EXPERIENCE

Graduate Assistant, Fordham University 08/2025 – Present | New York

- Analyzed 70k+ rows of data from the UNHP research group to create a model predicting distressed households with 98% accuracy.
- Built scalable data pipelines using Hugging Face, Keras, and Dynamic ML SMOTE to address class imbalance.
- Currently writing a research paper showcasing findings of socioeconomic insights discovered during the research.

Project Lead and Software Engineer, RateEat 01/2024 – 10/2024

- Led a cross-functional team of 9 members for a food review social app development.
- Worked on data analysis of 15k+ rows of large restaurant, food, and user activity data using Python scripts for cleaning, pre-processing, and storing.
- Implemented mobile app with API integration, authentication, session management, and data caching.

PROJECTS

Checkers bot Development, Reinforcement Learning, Search algorithm

- Used parallelism, iterative deepening, and bitboards and magic bitboards (advanced bitboard technique) to deepen search capabilities, which are essential for creating a competitive bot.
- Implemented the classical Minimax optimization, Alpha-Beta Pruning, and Zobrist Hashing to optimize decision-making, reducing computation time by 84% of the original computation in an average search depth.

Forest fire prediction, ML Model training

- Developed models on forest fire prediction using Scikit-learn, PyTorch, and Seaborn.
- Evaluated models, including neural networks, SVM, and the Rakel ensemble method, to find the best performing models.

Fare, FastAPI, Searching Algorithms

- Used Python Object-Relational Mapper (ORM) and SQL Alchemy for database migrations. Created the shortest path algorithm that takes time, cost, and other factors into consideration to provide users with multiple transportation options.
- Designed PostgreSQL schemas and secured backend with JWT authentication.

EDUCATION

M.S. Data Science, Fordham University 08/2025 – Present | New York, USA

- Pursuing a master's degree as a graduate researcher with the University Neighborhood Housing Program (UNHP) on housing issues in the Bronx borough.

B. Sc Software Engineering, Addis Ababa University ☑ 09/2017 – 06/2022 | CGPA: 3.5/4.0

- Graduated with a great distinction honor. ☑

COURSES

Supervised Machine Learning: Regression and Classification, Stanford Online ☑

UN Machine Learning Training, United Nations Frontier Tech Leaders Programme ☑

- Participated in the 2025 United Nations Development Program Machine Learning cohort.

Data structures and algorithms, A2SV ☑

- Completed a 1-year data structure and algorithm training program.
- Top 15% ☑ problem solver on LeetCode, solved 600+ ☑ questions across LeetCode and Codeforces.

SKILLS

Frameworks & Libraries: Python ML (Pandas, NumPy, Matplotlib, Seaborn), FastAPI, React.js, Flutter

Data Science: Regression & Classification, Clustering, Model Optimization, Learning Algorithms, Data Visualization

Technologies & Tools: Docker, Git, Jupyter, PostgreSQL, REST APIs, Kubernetes, Data Cleaning & ETL